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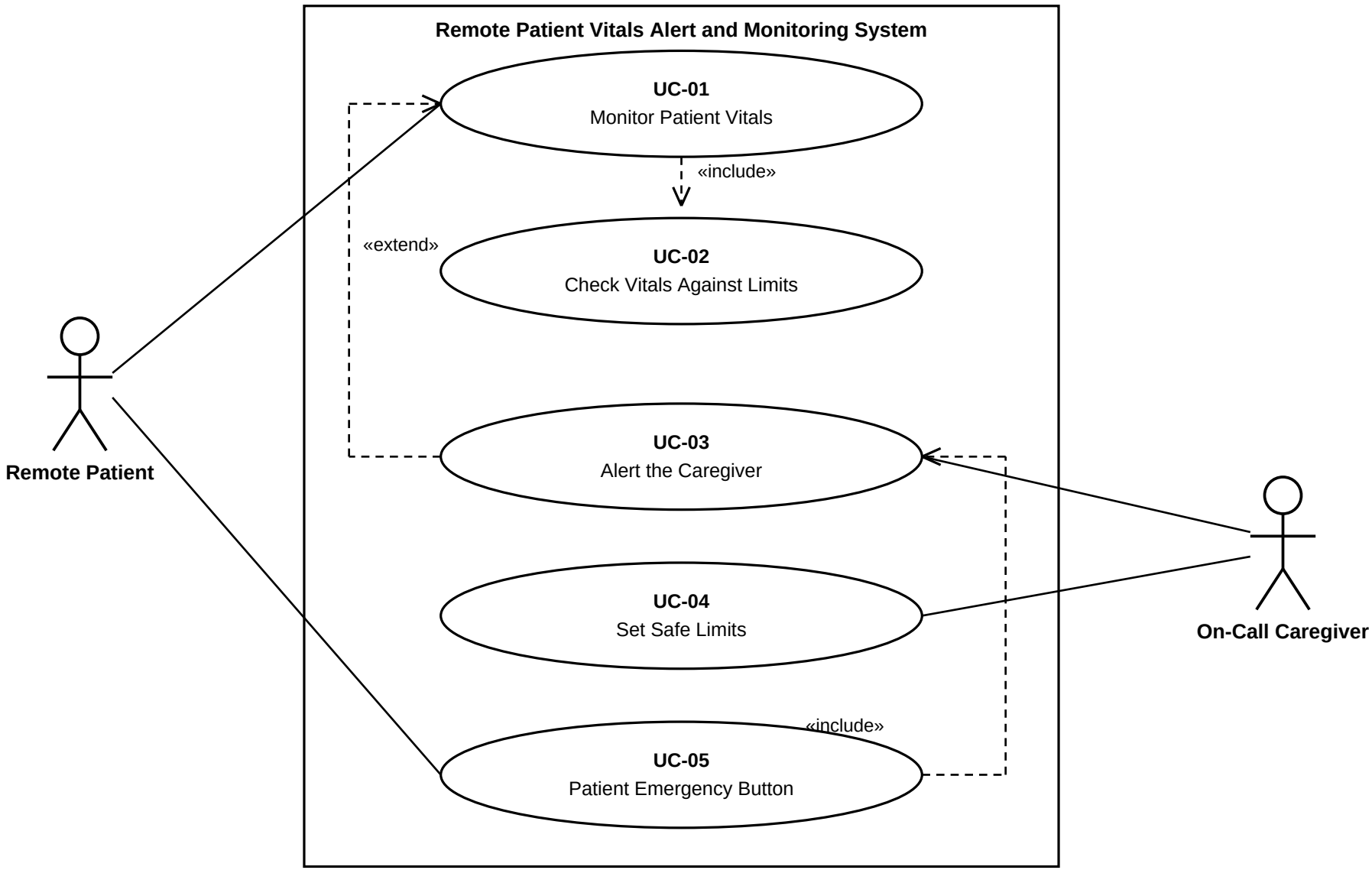
Part 1: Requirements Table

ID	Type	Description	Priority	How to test it	Why it matters
FR-001	Functional	The system shall collect the patient's SpO2, heart rate and blood pressure from their device at least once every 5 seconds.	High	Pass: new readings arrive at least every 5 seconds. Fail: a gap longer than 5 seconds with no error shown.	Nothing else in the app works without a steady stream of readings.
FR-002	Functional	The system shall check every reading against the patient's safe limits and raise an alert within 2 seconds if a limit is crossed.	High	Pass: a heart rate above the set limit raises an alert within 2 seconds. Fail: the alert takes more than 5 seconds, or never appears.	The whole point of the app is catching a problem fast.
FR-003	Functional	The system shall alert the first caregiver on the patient's contact list, and alert the next caregiver if that alert is not delivered within 60 seconds.	High	Pass: first caregiver is alerted, and if it does not go through the second is alerted by 60 seconds. Fail: the system stops after the first caregiver.	An alert must not be lost just because one caregiver is unreachable.
FR-004	Functional	The system shall let the on-call caregiver set the highest and lowest safe value for each vital sign, for each patient.	Medium	Pass: a saved limit is used on the very next reading. Fail: the saved limit is ignored, or an unsafe value is accepted.	Safe ranges are different for every patient after surgery.
FR-005	Functional	The system shall let the patient raise an emergency alert with one tap, even when their readings look normal.	Medium	Pass: one tap alerts the first caregiver within 10 seconds. Fail: no alert is sent, or more than one tap is needed.	A patient can feel very unwell while the three tracked vitals still look fine.
NFR-001	Nonfunctional (Speed)	The system shall handle 500 patients sending readings at the same time and still raise alerts in under 2 seconds for 95 out of every 100 alerts.	High	Pass: a load test with 500 patients meets this and loses no data. Fail: alerts are slower than 2 seconds, or any patient's data is dropped.	500 patients at once is the stated target load for the system.
NFR-002	Nonfunctional (Security)	The system shall scramble patient readings while being sent and while stored, and shall show a patient's records only to caregivers on that patient's contact list.	High	Pass: a network check finds no readable patient data, and an outsider is blocked. Fail: readable data is sent, or an outsider can open the records.	Health records are private and must stay with the right people.

Part 2: Use-Case Diagram



Part 3: Use-Case Flow

Use Case UC-03: Alert the Caregiver

Main actor: On-Call Caregiver, the person who receives the alert.

Other actor: Remote Patient, whose device sends the readings.

What starts it: A reading crosses one of the patient's safe limits.

Before it can run

1. The patient is registered and their device is connected and sending readings.
2. Safe limits have been set for this patient.
3. At least one caregiver is on the contact list with a working phone or email.

After it finishes

1. An alert is saved with the vital sign, its value, the limit crossed, and the time.
2. Either a caregiver has been reached, or the whole list has been tried and the failure is recorded.

Main flow (everything works)

1. The system receives a new reading from the patient's device.
2. The system checks it against the safe limits (this always runs UC-02).
3. The system finds the limit has been crossed and raises an alert within 2 seconds.
4. The system saves the alert with the vital sign, its value, the limit, and the time.
5. The system opens the patient's contact list and picks the first caregiver.
6. The system sends that caregiver the alert, showing who the patient is, which vital sign is wrong, and what the reading was.
7. The alert is confirmed as delivered within 60 seconds.
8. The system records the alert as delivered and notes which caregiver got it.
9. Done.

Alternate flow 7a: the first caregiver cannot be reached

7a1. No delivery confirmation comes back within 60 seconds of step 6.

7a2. The system picks the next caregiver on the contact list.

7a3. The system sends them the alert and goes back to step 7.

7a4. If the whole list has been tried and nobody was reached, the system records the failure and shows the alert on the patient's own screen so they know to seek help themselves.